
Geotechnical Investigation Report — Pier P4 Supplementary Pile Group

Nandira River Bridge Project — Package NRB-4

Document Reference:	GEO-NRB4-004
Date:	16-Nov-2022
Contract No.:	NHIA/NRB4/CW/2020-01
From:	Geotechnical Section, Meridian Engineering Consultants
To:	Contractor, Sagara Constructions Pvt. Ltd.
CC:	Employer, National Highways Infrastructure Authority (for information)

Subject: Geotechnical Investigation — Pile Position P4-N4 and Surrounding Pile Group, Pier P4

Further to Notice CTR-NRB4-0125 dated 09-Nov-2022, we report the findings of a supplementary geotechnical investigation carried out at pile position P4-N4 and the surrounding pile group between 10-Nov-2022 and 15-Nov-2022, comprising one rotary core borehole adjacent to P4-N4 and review of arisings retained from pile positions P4-N3, P4-N5 and P4-N6.

FINDINGS:

The borehole confirms a lens of soft, compressible clay between approximately 17.6m and 21.4m depth beneath pile position P4-N4, underlain by the dense sandy gravel stratum shown as the founding stratum in the geotechnical information provided with the tender documents. No equivalent lens was encountered in the arisings from pile positions P4-N3, P4-N5 or P4-N6, indicating the condition is localised to the immediate vicinity of P4-N4.

The geotechnical information provided with the tender documents does not show, or reasonably indicate the presence of, a compressible stratum in this vicinity. In our assessment, the condition encountered at pile position P4-N4 could not reasonably have been anticipated by an experienced contractor from that information.

RECOMMENDATION:

Pile position P4-N4 cannot achieve the design founding criteria at its originally specified socket depth. We recommend the pile socket at this position be extended by approximately 3.5m, founding within the underlying dense sandy gravel stratum, together with a corresponding increase in reinforcement cage length. A revised pile schedule for position P4-N4 is enclosed for the Engineer's instruction.

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